

Agile Software Assurance: An Empirical Study

Noura Abbas

*School of Electronics and Computer Science, University of Southampton
Southampton, SO17 1BJ, United Kingdom
n.abbas@ecs.soton.ac.uk*

Abstract

This poster will describe an empirical study of Agile Software Assurance. The main goal of this research is to study the quality of Agile projects in order to help software development organizations to have deeper understanding of Agile Methods, principles and practices. Moreover, this research will help evaluating the quality of Agile projects.

1. Research Questions

This Study will help to answer the following research questions:

1. What is the quality of Agile Software, importantly, how we can evaluate the quality of Agile Software?
2. Can Agile assure the quality under time pressure and with unstable requirements?
3. What are the best ways to assure the quality of Agile Software?
4. Different organizations interpret words like Agile and iterative in different ways. This leads to another question: how do organizations understand these terms?

2. Related Work

There have been many initiatives relating Agile with quality standards such as CMMI and ISO 9000. Although the results of these initiatives vary, they all shared the same idea of modifying Agile methods in order to be accepted from the standards point of view [1, 2, 3].

3. Methodology, Results and Future Work

Interviews and questionnaires constitute the appropriate qualitative and quantitative research

methods to collect as much data as possible about Agile in practice. We have started interviewing people from four organizations, and have talked with fifteen individuals who had experience with Agile projects. The interviewees are involved in different roles within the organizations. The projects were chosen from a range of software development organizations.

In addition, we have developed two questionnaires which will be used to evaluate 10 projects that use Agile. The first questionnaires will be to understand what organizations mean by the term “Agile”, and how Agile they are. Then the second one will be to find out how organizations assure the quality of Agile software products, and what are the reasons behind their success.

Our initial findings showed that the final products had different levels of quality. In addition, organizations use different Agile practices to assure the quality: testing, re-factoring, pair programming, recruiting skilled people, and continuous communication with the customer. Furthermore, the results showed that organizations prefer to adopt Agile practices rather than following one particular method. Importantly, those organizations have different understanding of Agility, and probably this is one of the important reasons behind having “Agile” software with different levels of quality.

The future work will be to analyze the collected data, in order to understand how people used Agile; and to find out how this usage affects the quality of the final products. The final stage will be to bring all the findings together and inform the model which will describe how to improve the quality of Agile projects.

4. References

- [1] Anderson, D. J. (2005). Stretching Agile to fit CMMI level 3 Microsoft Corporation
- [2] Nawrocki, J. R., M. Jasinski, et al. (2002). Combining XP with ISO 9000 . LNCS 2510 786-794.
- [3] Vriens, C. (2003). Certifying for CMM Level 2 and ISO9001 with XP@Scrum Proc. of (ADC'03)